Nolan Pierce

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Computer science student (Stanford, Visual Computing) pivoting toward hardware-software integration in robotics and manufacturing, with robotics/embedded systems coursework and Boeing experience in real-time 3D simulation.

Education

Stanford University — Stanford, CA
Bachelor of Science in Computer Science; GPA: 3.7 — *09/2023 – 06/2027*
Relevant Coursework: Introduction to Robotics (CS223A), Introduction to Electrical Engineering (ENGR40M), Computer Organizations and Systems (CS107), Deep Learning for Computer Vision (CS231N)

Technical Skills

Hardware & Robotics: Circuit prototyping & debugging (ENGR40M), robotics fundamentals – kinematics, control, sensor/actuator systems (CS223A), Arduino
Languages: Python, C, C++, SQL
Simulation & Graphics: Unreal Engine 5 (Blueprint scripting, event dispatchers, widget/UI systems)
Computer Vision & ML: PyTorch, OpenCV, Convolutional Neural Networks (CNNs), Vision Transformers (ViT)
Databases: PostgreSQL, Supabase (Auth/RLS), SQL
Frameworks & Tools: Node.js, Express.js, Git, Unix, gdb, REST APIs, JWT Authentication

Relevant Work Experience

Boeing Global Services — Seattle, WA
Systems Engineering Intern – Data & Analytics, VR Simulation & Modeling — *06/2026 – 09/2026*

- Engineered a virtual reality environment in Unreal Engine 5 to showcase employee-developed and patented safety and ergonomics solutions, extending a single-site physical "Dojo" training model into a **scalable, multi-site virtual training tool**
- Built interactive Blueprint systems – custom UI widgets, interface classes, event dispatchers, and parent/child blueprint hierarchies – to **streamline integration** of new training content into the existing simulation architecture
- Presented a live demo of the VR training environment to **Boeing leadership and cross-functional stakeholders** at OshKosh AirVenture
- Collaborated with engineering and safety teams to translate physical training workflows into an interactive 3D digital environment, bridging hardware-based training processes with software simulation

Mission Malama — Remote
Founding Engineer — *07/2025 – 09/2025*

- Designed and implemented backend architecture, including REST API endpoints, database schema, and CRUD operations using Node.js, Express.js, and Supabase (Postgres, Auth/RLS)
- Implemented JWT-based authentication and role-based access controls to ensure secure handling of sensitive user data
- Collaborated with founders to translate product vision into scalable, production-ready solutions

Washington Technology Student Association — Remote
Alumni Representative — *03/2023 – Present*

- Manage a network of 150+ alumni, facilitating mentorship programs, event planning, and volunteer recruitment

Candidate & Entertainment Manager — *03/2026 – Present*

- Mentor 5 candidate students and serve on the interview committee, narrowing 15 applicants to a Top 12
- Plan and execute entertainment logistics for a 2,000+ attendee state conference across 3 event nights

Personal Projects

Custom Heap Allocator | *C*

- Recreated dynamic memory allocation functions in C, demonstrating proficiency with pointers, memory management, and low-level systems programming

Multi-Agent Tracking & Trajectory Forecasting | *Python, PyTorch, OpenCV, Modal*

- Built an end-to-end perception-to-prediction pipeline for basketball video: SAM 3.1 multi-object player tracking (SportsMOT dataset) feeding a residual, rule-conditioned LSTM forecaster in PyTorch that predicts 4 future frames from an 8-frame observation window
- Matched a constant-velocity baseline (5.81 px median displacement error) and outperformed a plain LSTM on 10 of 12 test windows; trained and evaluated on GPU via Modal, validating transfer across held-out video clips

Leadership & Extracurriculars

Stanford Men's Rugby — Stanford, CA
President, Vice President — *05/2025 – Present*

- Led 40+ team members and managed administrative operations for the club